

Worksheet 10

October 7, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

1. Composition and domain

Let $f(x) = \frac{1}{x-1}$ and $g(x) = \sqrt{x+2}$.

(a) Find $(f \circ g)(x)$.

(b) State the domain in interval notation. Keep all restrictions from both functions.

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2. An inverse function

Let $h(x) = 3x - 5$.

(a) Find $h^{-1}(x)$ by interchanging x and y and solving for y .

(b) Check your answer by simplifying $h(h^{-1}(x))$.